

Global Energy Storage

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America

[R3 Lithium launches lithium recycling facility in Georgia.- electrive.com](#) R3 Lithium has begun commercial production of battery-grade lithium carbonate from recycled battery materials at its facility in Covington, Georgia. The plant, acquired from bankrupt recycler Ascend Elements in July 2026, is designed to produce 2,500 tonnes of lithium carbonate annually and process up to 30,000 tonnes of battery scrap and end-of-life batteries into black mass. R3 plans to invest an additional US\$15 million to support continuous commercial operations and has already secured approximately US\$1 billion of offtake agreements, including with commodity trader Trafigura. The facility represents one of the first commercial-scale lithium recycling operations in North America and supports efforts to localize critical battery material supply chains in the US.

[Washington seeks to sever Ford's Ties with Chinese partners.- electrive.com](#) Ford is facing renewed scrutiny from the US government over its partnerships with Chinese companies, particularly CATL and Geely. US Transportation Secretary Sean Duffy urged Ford to reconsider its reliance on CATL technology, which will be used at Ford's BlueOval Battery Park Michigan to manufacture LFP batteries for future EVs. Ford defended the arrangement, emphasizing that the Michigan facility is fully owned and operated by Ford, with CATL providing only technology licensing and technical support rather than direct ownership. The administration also raised concerns about Ford's expanding cooperation with Geely in Europe, where the companies plan to jointly develop EV models and manufacture Geely vehicles at Ford's Valencia, Spain plant starting in 2028. The dispute highlights growing political pressure on US automakers to reduce dependence on Chinese battery technology despite ongoing efforts to localize battery production in North America.

[LG Energy Solution reports progress in LMR batteries.- electrive.com](#) LG Energy Solution (LGES) and Seoul National University have developed a new approach to improve the stability of LMR (lithium manganese-rich) batteries, a next-generation chemistry that reduces reliance on cobalt while offering high energy density. By optimizing charging/discharging voltage ranges and the cell formation process, researchers increased oxygen recovery from 86% to 97%, reducing gas generation and structural degradation. In testing, optimized 40 Ah LMR cells retained 92.2% of their original energy after 883 cycles, demonstrating significantly improved cycle life. The results support LGES' efforts to commercialize lower-cost, high-manganese battery technologies for future EV and energy storage applications, although no commercialization timeline has been disclosed.

['Deeply Alarmed': Trump Administration Slams Ford's Deal With Chinese Battery Company.- insideevs.com](#) US Transportation Secretary Sean Duffy urged Ford to reconsider its partnerships with Chinese companies, specifically criticizing Ford's battery technology licensing agreement with CATL and its cooperation with Geely. The focus of the criticism is Ford's BlueOval Battery Park Michigan, where CATL technology is being used to manufacture LFP batteries for future Ford EVs, including the upcoming US\$30,000 Fathom electric pickup. Ford defended the arrangement, emphasizing that the facility is fully owned and operated by Ford and that the CATL agreement is limited to technology licensing rather than a joint venture. The dispute highlights growing political pressure in Washington to reduce US reliance on Chinese EV and battery supply chains, despite the industry's continued need for Chinese battery technology to support cost-competitive EV production.

Asia

[China Targets 70% NEV Sales in Passenger Vehicle Market by 2030.- yicaiglobal.com](#) China has released a new 2026-2030 development plan for intelligent connected new energy vehicles (NEVs), targeting 70% NEV penetration in passenger vehicle sales and 40% penetration

in commercial vehicle sales by 2030. The plan also aims for the automotive industry to reach peak carbon emissions by 2030 and accelerate supply chain decarbonization. A key focus is tighter oversight of excess manufacturing capacity in both the automotive and power battery sectors, with authorities promoting market-driven consolidation, restructuring, and the phase-out of inefficient production capacity to improve utilization rates and reduce destructive price competition. The roadmap further calls for large-scale deployment of autonomous driving technologies by 2030 and reiterates support for foreign investment and international cooperation in China's EV and battery industries.

[China's Energy Storage Cell Sector Is Set for Years of Capacity Clearing After Boom, Sources Say.- yicaiglobal.com](#) China's energy storage battery cell industry is expected to undergo several years of capacity rationalization and consolidation as regulators move to address growing structural overcapacity. Despite energy storage battery production exceeding 400 GWh in 1H26 and new storage project tenders surpassing 100 GWh, total industry capacity is projected to reach 1.2-1.5 TWh by the end of 2026, with more than 2 TWh of completed, under-construction, and planned capacity. Industry sources expect outdated production lines, particularly those focused on sub-314 Ah battery cells, to be gradually phased out, while weaker manufacturers may exit the market. The policy shift aims to improve capacity utilization, curb excessive price competition, strengthen industry standards, and better align future battery production with actual demand growth in China's rapidly expanding energy storage sector.

[China's Yuneng to Likely Discount Hong Kong Listing Price as Top Shareholders Cut Holdings, Insider Says.- yicaiglobal.com](#) Hunan Yuneng New Energy Battery Material, the world's largest supplier of LFP cathode materials, may need to price its planned Hong Kong IPO at a discount to its mainland-listed shares despite reporting strong financial results. Investor sentiment has been weakened by significant share sales from major shareholders, including CATL, which plans to reduce its stake by up to 3%, alongside other early investors. This comes even as Yuneng reported 1H26 revenue of CNY 34.9 billion (+143% YoY) and net profit of CNY 2.9 billion (+854% YoY), driven by strong demand for LFP cathode materials. The company sold 667,200 tonnes of cathode materials in 1H26, with energy storage applications accounting for 53% of total sales, and maintained its leading position with a 28% global LFP market share and 55% share in the high-end LFP segment. The IPO proceeds are intended to support an integrated battery materials project in Guizhou covering mining, lithium processing, and recycling.

[China's Li Auto to Inject USD402.3 Million to Raise Stake in Battery Maker Sunwoda EVB.- yicaiglobal.com](#) Li Auto plans to invest CNY 2.7 billion (US\$402 million) in Sunwoda EVB, increasing its ownership stake to 8.8% and becoming the company's second-largest shareholder. Following the transaction, Li Auto-related entities will hold a combined 11.2% stake in the battery manufacturer. The investment is intended to strengthen cooperation in battery R&D, manufacturing, quality control, and next-generation battery development, while enhancing supply security and delivery reliability. The two companies have worked together since 2017, and Sunwoda currently serves as the contract manufacturer for Li Auto's self-developed 5C fast-charging batteries. The move further deepens vertical integration efforts as Li Auto accelerates the development of its in-house battery technology capabilities.

[Hyundai Motor Group Partners on Used EV Battery ESS Project.- thelec.net](#) Hyundai Motor Group, LG Energy Solution (LGES), Hyundai Engineering, and Wonik PNE have launched South Korea's first demonstration project using retired E-GMP EV batteries in an energy storage system (ESS). The project will deploy a 200 kWh used-battery ESS integrated with an EV fast-charging station, storing electricity during off-peak hours and supplying power during peak periods to improve charging economics and grid efficiency. LGES will be responsible for battery diagnostics, safety verification, and ESS construction, while Hyundai Engineering will assess commercial feasibility and integration with charging infrastructure. The initiative is aimed at validating the technical performance, safety, and business case for second-life EV battery applications and supporting the development of a battery reuse ecosystem in Korea.

[CIA-Backed BEI to Expand Battery Capacity Eightfoldn.- thelec.net](#) South Korean battery startup BEI plans to expand annual production capacity from 24.3 MWh to 200 MWh by 2027, supported by KRW 30 billion in newly raised Series B funding and growing demand from the drone, defense, and robotics sectors. The company's proprietary anode-free lithium-metal battery technology eliminates graphite and lithium-metal foil, enabling higher energy density, lower manufacturing costs, and improved cold-weather performance. BEI has attracted investment from In-Q-Tel (IQT), the venture arm backed by the US intelligence community, and already has signed letters of intent covering 7.64 million cells, exceeding two years of planned production. The company is targeting further global expansion, a KRW 50 billion Series C funding round in 2027, and an IPO in 2028.

Europe

[End after 700 e-buses: Bolloré pulls the plug on Bluebus- electrive.com](#) Bolloré has decided to discontinue electric bus production at its subsidiary Bluebus after delivering nearly 700 electric buses to around 60 public transport operators, primarily in France. The company said Bluebus failed to achieve the scale needed to compete effectively in the increasingly competitive electric bus market. The business was also impacted by battery-related fire incidents in Paris that led to the temporary grounding of 232 Bluebus vehicles. While bus manufacturing will be phased out, Bolloré will continue investing in its battery subsidiary Blue Solutions, focusing on its next-generation Gen4 solid-state batteries, which target <20-minute charging times, 40% higher range, and improved safety. Blue Solutions aims to commercialize the technology and establish a gigafactory by 2029, supported by a strategic investor.

[Aviloo expands battery warranty to 26 countries.- electrive.com](#) Aviloo has expanded its used EV battery warranty program to 26 European countries, with coverage automatically included for vehicles that pass the company's Flash Test battery diagnostic assessment. The warranty is linked to Aviloo's battery health evaluation, which determines the battery's State of Health (SoH) through the vehicle's OBD interface. The expansion follows Aviloo's analysis of more than 500,000 battery tests across 20 EV models, which showed most vehicles retained 87-94% battery health after 150,000 km. The initiative aims to reduce uncertainty in the used EV market by providing independent battery condition verification and additional warranty protection for buyers, dealers, and leasing companies.

[IBU-tec secures key approval for LFP production plant in Germany.- electrive.com](#) IBU-tec, a German battery materials supplier and partner of Volkswagen's PowerCo, has received a key regulatory approval for its planned LFP cathode material plant in Bitterfeld-Wolfen, Germany. The facility is designed to produce 15,000 tonnes of LFP cathode material annually and is scheduled to begin operations in 2028. Under a previously signed agreement, PowerCo has secured 100% of the plant's output for 10 years, providing IBU-tec with a long-term revenue stream. The company expects to become the first industrial-scale producer of LFP cathode material in Europe, supporting regional battery supply chains for EVs, energy storage systems, and other strategic applications.

[Lanxess commissions battery laboratory for LFP materials in Germany.- electrive.com](#) Lanxess has opened a new battery research laboratory at its Krefeld-Uerdingen, Germany site to accelerate the development of materials for LFP (lithium iron phosphate) batteries. The facility will enable the company to directly evaluate how its iron oxide and iron phosphate precursor materials perform inside battery cells, helping shorten development cycles and improve qualification of next-generation battery materials. Beyond LFP batteries, the lab can also support research into sodium-ion battery materials. The investment strengthens Lanxess' position in Europe's growing battery supply chain and supports the company's ambition to become a key supplier of battery materials as LFP adoption expands across EV and energy storage markets.

[Battery storage startup Rebaba secures nearly €4 million.- electrive.com](#) Swedish energy storage startup Rebaba, founded by former Northvolt employees, has raised SEK 44.1 million (€4 million) in an oversubscribed seed funding round to expand its second-life battery storage business. The company repurposes retired EV batteries, which typically retain 70-80% of their usable capacity, into stationary energy storage systems for residential, commercial, and industrial applications. Rebaba plans to expand its Stockholm production facility to 40 MWh of annual capacity, open additional manufacturing sites across Europe, and accelerate deployment of its circular battery storage solutions. The funding supports growing demand for energy storage driven by renewable energy integration, electrification, and AI-related power demand while promoting a more circular battery value chain.

[Fraunhofer ISE increases energy density of battery cells.- electrive.com](#) Fraunhofer ISE has developed a new battery cell architecture that increases energy density by 10-15% without increasing cell weight by using significantly thicker electrode coatings (up to 800 µm vs. the typical 100-200 µm). The design reduces the number of current collectors required, creating more space for energy-storing active materials within the cell. Initially developed for battery energy storage systems (BESS), the technology could also benefit electric vehicles by enabling longer driving ranges. While not yet ready for commercial production, the research demonstrates a promising pathway for future battery manufacturing, particularly for next-generation sodium-ion and zinc-ion battery applications.

[Toptec Wins 25.6 Billion Won Hyundai Mobis Battery Equipment Order in Czech Republic.- thelec.net](#) Toptec has secured a KRW 25.6 billion (US\$19 million) contract from Hyundai Mobis' Czech subsidiary to supply a pouch-type EV battery assembly line, with delivery scheduled through April 2027. The contract represents about 15% of Toptec's 2025 revenue. Combined with two recent Hyundai Mobis orders for HEV and EREV battery assembly lines worth KRW 20.1 billion and KRW 21.1 billion, respectively, Toptec has secured a total of KRW 66.8 billion in battery equipment orders from Hyundai Mobis within two weeks, equivalent to roughly 39% of its annual revenue. The orders reflect Hyundai Mobis' ongoing expansion of EV, hybrid and EREV battery production capacity in both Europe and North America.

[A Mercedes-Backed Company Just Put Solid-State Batteries Into Mass Production.- insideevs.com](#) ProLogium, a Taiwanese battery developer backed by Mercedes-Benz, has begun mass production of its Gen 3.5 all-solid-state battery cells at its Taoyuan facility, marking one of the industry's first commercial-scale solid-state battery production programs. The large-format 185.4 Ah pouch cell delivers an energy density of 381 Wh/kg and 903 Wh/L, around 30% higher than conventional NMC batteries. The company says its technology supports 5%-80% charging in under 10 minutes while offering enhanced safety through a composite solid electrolyte and ceramic separator design. Initial production capacity is 0.5 GWh, with plans to expand capacity in Taiwan and construct a new 4 GWh facility in France, targeting commercial scale-up through 2030.

Others

[Denmark's CIP Buys Australian Battery and Wind Farm Project.- BNEF](#) Copenhagen Infrastructure Partners (CIP) has acquired full ownership of the Gawara Baya renewable energy project in Queensland, Australia, from Windlab and reached a final investment decision. The project combines a 408 MW onshore wind farm with a 104 MW battery energy storage system (BESS) and is expected to be fully operational by 2030. CIP has secured A\$1.7 billion (US\$1.2 billion) in financing from 10 banks to support construction. Backed by Australia's Capacity Investment Scheme, the project is expected to generate enough electricity to power approximately 240,000 homes and supports Australia's transition

away from coal-fired generation. The investment further expands CIP's renewable energy portfolio across the Asia-Pacific region.

[This Study Covered 500,000 EV Batteries. Some Models Aged Much Better Than Others.- insideevs.com](#) Battery diagnostics firm Aviloo analyzed 500,000 EV battery health tests across 20 popular EV models and found that most vehicles retain more than 90% of their original battery capacity after 150,000 km (93,000 miles). The Hyundai Ioniq 5 was the top performer, retaining 97% capacity at 50,000 km, 95% at 100,000 km, and 93.8% at 150,000 km. Other strong performers included the Mercedes EQA, Hyundai Kona Electric, BMW i4, and Ford Mustang Mach-E. Tesla's Model Y and Model 3 retained 90.3% and 88.9% capacity, respectively, after 150,000 km. Battery degradation was generally higher in older air-cooled models such as the Nissan Leaf and Renault Zoe, while factors such as climate, charging behavior, and battery chemistry significantly influenced long-term battery health.

EXHIBIT 1: Key commodities price performance

	Price 11-Sep	Ccy	Mkt cap \$bn	P/E 26E	EV /EBITDA 26E	%1D	%1W	%1M	%1Y	1Y px chart
Metal prices (US\$/tonne)										
LiCO (spot)	20,946					-2.0%	-10%	1%	109%	
LiOH (spot)	19,455					-2.2%	-10%	1%	87%	
LiCO (contract)	18,000					0.0%	0%	-10%	89%	
LiOH (contract)	18,500					0.0%	0%	-5%	113%	
Cobalt (spot)	41,950					-1.3%	-9%	-25%	27%	
Nickel (spot)	16,666					-1.4%	-1%	-1%	10%	
Component prices										
Cathode NMC811						-0.8%	-3%	-6%	27%	
Cathode LFP						-1.6%	-5%	-3%	52%	
Precursor NMC811						-0.2%	-2%	-9%	9%	
Precursor LFP						0.0%	1%	2%	52%	
Artificial Graphite						0.0%	0%	0%	6%	
Natural Graphite						0.0%	0%	0%	0%	
Electrolyte NMC						0.0%	0%	2%	38%	
Electrolyte LFP						0.0%	0%	3%	59%	
Separator						0.0%	0%	0%	28%	
Spot battery cell cost (US\$/kWh)										
LFP	60					-0.4%	-1%	0%	15%	
NMC532	82					-0.6%	-3%	-4%	17%	
NMC622	78					-0.6%	-2%	-4%	12%	
NMC811	81					-0.4%	-2%	-3%	15%	
Spot battery pack cost (US\$/kWh)										
LFP	73					-0.3%	-1%	0%	-3%	
NMC532	113					-0.4%	-2%	-3%	12%	
NMC622	111					-0.4%	-2%	-3%	8%	
NMC811	114					-0.3%	-1%	-2%	10%	

Source: Bloomberg, Baiinfo, Bernstein analysis and estimates (cell and pack cost)

EXHIBIT 2: Key companies price performance and valuation

	Price 11-Sep	Ccy	Mkt cap \$bn	P/E 26E	EV /EBITDA 26E	%1D	%1W	%1M	%1Y	1Y px chart
Cell makers										
LGES	360,000	KRW	62.9	n.a.	15.2x	-1.1%	0%	1%	3%	
Samsung SDI	548,000	KRW	33.0	79.3x	16.7x	-2.3%	-1%	18%	176%	
CATL	330.5	CNY	232.5	15.9x	11.1x	-2.2%	-6%	-15%	2%	
CALB	16.9	HKD	3.8	9.5x	2.5x	-0.9%	-1%	-9%	-38%	
EVE Energy	48.6	CNY	15.7	14.4x	9.1x	-2.5%	-5%	-13%	-32%	
Gotion	25.0	CNY	6.8	23.5x	5.8x	-1.8%	-4%	-10%	-47%	
Sunwoda	18.5	CNY	5.1	17.0x	5.4x	3.4%	10%	-1%	-34%	
Farasis	8.6	CNY	1.6	n.a.	n.a.	-2.5%	-5%	-15%	-58%	
Lithium Miners/Refiners										
Albemarle	117.6	USD	13.9	10.2x	5.0x	-3.7%	-7%	-9%	57%	
SQM	71.2	USD	20.1	9.5x	4.7x	-1.8%	-7%	-1%	67%	
Pilbara Minerals	4.5	AUD	10.5	24.8x	12.7x	-7.4%	-11%	-3%	129%	
Ganfeng Lithium	34.7	HKD	13.4	7.8x	7.0x	-3.4%	-5%	-13%	2%	
Tianqi Lithium	43.0	CNY	10.4	10.1x	3.7x	-3.8%	-5%	-11%	-2%	
Cathode										
LG Chem	272,500	KRW	14.4	n.a.	2.6x	-3.0%	-4%	-1%	-6%	
EcoPro BM	110,900	KRW	8.3	561.7x	54.7x	-6.2%	4%	3%	-3%	
L&F	120,800	KRW	3.7	n.a.	15.9x	-1.8%	-2%	22%	91%	
Umicore	22.5	EUR	6.4	12.4x	5.5x	1.6%	-2%	-6%	70%	
Ningbo Ronbay	22.9	CNY	2.4	40.9x	11.0x	-4.5%	-7%	-18%	-21%	
Beijing Easpring	36.7	CNY	3.0	17.9x	11.6x	-2.8%	-5%	-13%	-34%	
GEM	6.2	CNY	4.7	16.7x	5.2x	-1.6%	-3%	-8%	-16%	
Huayou Cobalt	36.1	CNY	10.2	8.8x	3.8x	-3.6%	-5%	-17%	-30%	
Dynanonic	39.5	CNY	1.7	13.0x	5.5x	-3.4%	-5%	-15%	-4%	
Anode										
Posco	186,200	KRW	12.4	484.8x	48.8x	-2.9%	1%	19%	42%	
Putailai	21.4	CNY	6.8	14.1x	8.8x	-2.7%	-5%	-12%	-20%	
Shanshan	11.2	CNY	3.7	n.a.	n.a.	-2.4%	-3%	-8%	-17%	
Separator										
Asahi Kasei	1,552	JPY	13.8	14.3x	5.4x	-0.9%	-4%	-8%	29%	
SK IE	16,680	KRW	1.0	n.a.	-20.8x	-6.9%	2%	9%	-40%	
Toray Industries	1,237	JPY	12.1	23.0x	7.4x	0.7%	-1%	-10%	26%	
W-Scope	149	JPY	0.1	n.a.	n.a.	-8.6%	-12%	-6%	-39%	
Yunnan Energy	47.2	CNY	6.9	20.8x	8.0x	-2.9%	-2%	-15%	4%	
Shenzhen Senior	14.4	CNY	3.4	29.7x	10.4x	-2.3%	3%	4%	-1%	
Sinoma Science	51.5	CNY	12.9	28.8x	12.6x	1.3%	9%	-9%	42%	
Electrolyte										
Enchem	22,150	KRW	0.4	n.a.	n.a.	-3.7%	6%	-14%	-68%	
Tinci Minerals	32.3	CNY	9.8	10.2x	7.4x	-2.5%	-4%	-18%	13%	
Capchem	71.5	CNY	8.0	24.1x	18.0x	2.6%	14%	9%	47%	
Jiangsu Guotai	7.5	CNY	1.8	9.5x	n.a.	-1.3%	-2%	-2%	-15%	
Shanshan	11.2	CNY	3.7	n.a.	n.a.	-2.4%	-3%	-8%	-17%	
Equipment Maker										
Hirano	1,763	JPY	0.2	16.7x	n.a.	-1.6%	-6%	4%	8%	
CKD	5,060	JPY	2.2	27.2x	14.6x	-2.5%	-3%	-16%	100%	
Wuxi Lead	29.7	CNY	7.3	20.1x	16.1x	-1.3%	-6%	-15%	-51%	
Shenzhen Yinghe	20.1	CNY	1.9	19.5x	11.7x	-2.9%	-7%	-1%	-28%	
Zhejiang HangKe	20.0	CNY	1.8	22.7x	18.6x	0.7%	-2%	-10%	-50%	
Guangdong Lyric	30.9	CNY	0.8	33.4x	13.6x	-2.9%	-5%	-16%	-59%	
Shenzhen United	20.0	CNY	1.0	20.3x	12.5x	-2.0%	-5%	2%	-28%	

Source: Bloomberg (all estimates), Bernstein analysis
GLOBAL ENERGY STORAGE

Company Models

[CATL](#) | [LG Energy Solution](#) | [Samsung SDI](#)

[LG Chem](#) | [EcoPro BM](#) | [Posco Future M](#)

[Tianqi Lithium](#)

[Sungrow](#)

Industry Models

[Model: ESS tracker](#)

[Model: Monthly EV Tracker](#)

[Model: Battery Gigafactory Database](#)

[Industry model: Global EV TAM](#)

[Industry model: Global ESS TAM](#)

[Industry model: Battery TAM](#)

[Industry model: Battery supply and demand](#)

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BERNSTEIN TICKER TABLE

Ticker	Rating	Cur	11 Sep 2026		TTM	Reported EPS				Reported P/E (x)		
			Closing Price	Price Target	Rel. Perf.	Cur	2025A	2026E	2027E	2025A	2026E	2027E
300750.CH (CATL)	O	CNY	330.51	800.00	(21.6)%	CNY	16.14	21.95	28.77	20.5	15.1	11.5
3750.HK (CATL)	O	HKD	547.50	770.00	3.6%	CNY	16.14	21.95	28.77	29.0	21.3	16.3
051910.KS (LG Chem)	M	KRW	272,500	312,000	(31.2)%	KRW (13,258.70)	2,693.22	28,743		(20.6)	101.2	9.5
373220.KS (LGES)	M	KRW	360,000	351,000	(22.3)%	KRW (4,580.94)	(2,364.33)	10,225		(78.6)	(152.3)	35.2
006400.KS (SDI)	M	KRW	548,000	520,000	151.2%	KRW (9,933.80)	1,751.66	18,577		(55.2)	312.8	29.5
247540.KS (EcoPro BM)	U	KRW	110,900	140,000	(27.7)%	KRW	403.00	854.00	1,963.00	275.2	129.9	56.5
300274.CH (Sungrow)	O	RMB	84.40	151.00	(63.7)%	RMB	6.55	6.93	7.95	12.9	12.2	10.6
002466.CH (Tianqi Lithium)	O	CNY	43.00	80.00	(27.3)%	CNY	0.28	3.90	7.75	152.4	11.0	5.6
9696.HK (Tianqi Lithium)	O	HKD	31.82	65.00	(45.9)%	CNY	0.28	3.90	7.75	96.5	7.0	3.5
ASIA			1,962.10									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

Source: Bloomberg, Bernstein estimates and analysis.

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